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**EX-503 (GS)**  
**B.E. V Semester Examination, June 2020**  
**Grading System (GS)**  
**Electrical Machine - II**  
**Time : Three Hours**

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. Why is it necessary to insert resistance in the armature circuit of d.c. motors at the time of starting? Describe with a neat sketch the working principle of a shunt motor starter and explain how its safety devices work?
2. A synchronous motor is supplying a certain load in order to stop the motor, it is essential to Switch Off first the a.c. supply and then its field current. Explain the reason.
3. With the help of a neat diagram, describe the main parts of an alternator with their function.
4. A cylindrical rotor alternator is operating at a lagging power factor. Show that  $E I_a \cos \delta$  is  $\frac{EV}{X} \sin \delta$  where  $\phi$  is internal power factor angle and  $\delta$  is the load angle. Neglect armature circuit resistance.
5. Write short notes (any two) :
  - a) Reluctance motor
  - b) Stepper motor
  - c) Brushless DC motor
  - d) Swinburne's test
6. A D.C. generator operating at rated load has a terminal voltage of 240V. Determine the no load voltage if the voltage regulation is 2.5%.
7. Explain the causes of Harmonics in the voltage wave of the alternator. Discuss the methods of eliminating or reducing these harmonics.

OR

What are the function of interpole and compensating winding in D.C. machine?

8. Explain the following, (any two)
  - i) Starting of synchronous motor
  - ii) V curves
  - iii) Hunting and damper winding
  - iv) Zero power factor method

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